

PROJECT 1: GENDER GAP ANALYSIS

INITIAL INSPECTION

1. How many companies are in the data set?

Answer: 10,174 records

--Check distinct count of employername, companynumber

```
SELECT DISTINCT employername, companynumber  
FROM gender_pay_gap_21_22
```

	employername character varying	companynumber character varying
1	B & M RETAIL LIMITED	01357507
2	T.J.SMITH AND NEPHEW,LIMITED	00093994
3	SOCOTEC UK LIMITED	02880501
4	MYTIME ACTIVE	04809606
5	MARKS & CLERK LLP	0C343273
6	CYNERGY BANK LIMITED	04728421
7	PAUL SMITH LIMITED	01170719
8	Jones Day	0
9	TRESCAL LIMITED	06614164
10	SUPPORT FOR ORDINARY LIVING	SC183408
11	XBRIDGE LIMITED	03967717
12	WATSON FARLEY & WILLIAMS LLP	0C312252
13	E.ON CONTROL SOLUTIONS LIMITED	04681451
14	NICOVENTURES RETAIL (UK) LIMITED	10235033
15	CRM STUDENTS LIMITED	04886412
Total rows: 10174 Query complete 00:00:00.605		

```
SELECT COUNT(DISTINCT (employername, companynumber)) AS  
unique_number_of_companies  
FROM gender_pay_gap_21_22
```

	unique_number_of_companies bigint
1	10174

/*Additional verification queries*/

--Check distinct count of companynumber

```
SELECT COUNT(DISTINCT companynumber) AS  
unique_number_of_companynumber  
FROM gender_pay_gap_21_22
```

	unique_number_of_companynumber bigint
1	8810

--Check for same companynumbers

```
SELECT companynumber, COUNT(1) AS same_companynumbers
FROM gender_pay_gap_21_22
GROUP BY 1
HAVING COUNT(1) > 1
```

	companynumber character varying	same_companynumbers bigint
1	0	1365

--View the dataset with same companynumbers

```
SELECT *
FROM gender_pay_gap_21_22
WHERE companynumber = '0'
ORDER BY employername
```

	employername character varying	employerid character varying	address character varying
1	Aberdein Considine	16349	5-9 Bon Accord Crescent, Aberdeen, AB11 6DN
2	Abingdon & Witney College	799	Holloway Road, Witney, Oxfordshire, United Kingdom, OX28 6NE
3	Abraham Moss Community School	811	Crescent Road, Crumpsall, Manchester, M8 5UF
4	Accord Housing Association Ltd	15369	178 Birmingham Road, West Bromwich, West Midlands, B706QG
5	Activate Learning	17170	Oxpens Road, Oxford, OX1 1SA
6	Activate Learning Education Trust	18548	Oxpens Road, Oxford, OX1 1SA
7	Adrian Flux Insurance Services	17234	East Winch Hall, East Winch, Kings Lynn, PE32 1HN
8	Advanced Forwarding Limited	20576	606 House, Wharfedale Road, Euroway Trading Estate, Bradford, West Yorkshire, BD4 6SG
9	Advanced Supply Chain (BFD) Limited	20577	606 House, Wharfedale Road, Euroway Trading Estate, Bradford, West Yorkshire, BD4 6SG
10	Advanced Supply Chain (COR) Limited	20580	606 House, Wharfedale Road, Euroway Trading Estate, Bradford, West Yorkshire, BD4 6SG
11	Advanced Supply Chain Group Limited	20575	606 House, Wharfedale Road, Euroway Trading Estate, Bradford, West Yorkshire, BD4 6SG
12	Advanced Supply Chain (HF) Limited	20579	606 House, Wharfedale Road, Euroway Trading Estate, Bradford, West Yorkshire, BD4 6SG
13	Advanced Supply Chain (Logistics) Limited	20578	606 House, Wharfedale Road, Euroway Trading Estate, Bradford, West Yorkshire, BD4 6SG
14	Advisory, Conciliation and Arbitration Service	496	Euston Tower, London, London, United Kingdom, NW1 3JJ
Total rows: 1365 Query complete 00:00:00.765			

--Check distinct count of employername

```
SELECT COUNT(DISTINCT employername) AS unique_employername
FROM gender_pay_gap_21_22
```

	unique_employername bigint
1	10174

--Check for any NULL values of employername

```
SELECT *
FROM gender_pay_gap_21_22
WHERE employername IS NULL
```

employername character varying	employerid character varying	address character varying	postcode character varying	companynumber character varying	slcodes character varying	diffmeanhourlypercent numeric	diffmedianhourlypercent numeric	diffmeanbonuspercent numeric	diffmedianbonuspercent numeric	malebonus numeric
Total rows: 0 Query complete 00:00:00.643										
										LF Ln 46, Col 1

2. How many of them submitted their data after the reporting deadline?

Answer: 361 companies submitted after the reporting deadline
submittedafterthedeathline=true:361 records

```
SELECT submittedafterthedeathline,  
       COUNT(1) as number_of_companies  
FROM gender_pay_gap_21_22  
GROUP BY submittedafterthedeathline
```

	submittedafterthedeathline boolean	number_of_companies bigint
1	false	9813
2	true	361

3. How many companies have not provided a URL?

Answer: 3,700 companies

```
SELECT CASE  
       WHEN companylinktopginfo ILIKE 'HTTP%' THEN 'Has a URL'  
       ELSE 'No URL'  
END AS url_tag,  
       COUNT(1) AS number_of_companies  
FROM gender_pay_gap_21_22  
GROUP BY url_tag
```

	url_tag text	number_of_companies bigint
1	No URL	3700
2	Has a URL	6474

*/*Additional verification queries*/*

--Check possible values of the URL

```
SELECT DISTINCT LEFT(companylinktopginfo, 6) AS startingletters,  
       COUNT(1) AS count_url_startingletters  
FROM gender_pay_gap_21_22  
GROUP BY 1  
ORDER BY 1
```

	startingletters text	count_url_startingletters bigint
1	0	3700
2	http:/	270
3	Http:/	1
4	HTTP:/	1
5	https:	6199
6	Https:	2
7	HTTPS:	1

Total rows: 7 Query complete 00:00:00.625

```
SELECT COUNT(1) AS companies_with_URL
FROM gender_pay_gap_21_22
WHERE companylinktogpinfo ILIKE 'http%'
```

	companies_with_url
	bigint
1	6474

4. Which measures of pay gap contain too much missing data, and should not be used in our analysis?

Answer: Upon checking for all these measures for NULL values, all company records have a numerical value for each of these measures. Therefore, any of these measures can be used for our analysis.

- diffmeanhourlypercent
- diffmedianhourlypercent
- diffmeanbonuspercent
- diffmedianbonuspercent
- malebonuspercent
- femalebonuspercent
- malelowerquartile
- femalelowerquartile
- malelowermiddlequartile
- femalelowermiddlequartile
- maleuppermiddlequartile
- femaleuppermiddlequartile
- maletopquartile
- femaletopquartile

--Check for diffmeanhourlypercent

```
SELECT CASE
      WHEN diffmeanhourlypercent IS NULL THEN 'Missing Data'
      ELSE 'Has data'
    END AS diffmeanhourlypercent_tag,
      COUNT(1) AS number_of_companies
FROM gender_pay_gap_21_22
GROUP BY 1
```

	diffmeanhourlypercent_tag	number_of_companies
	text	bigint
1	Has data	10174

--Check for diffmedianhourlypercent

```
SELECT CASE
    WHEN diffmedianhourlypercent IS NULL THEN 'Missing Data'
    ELSE 'Has data'
END AS diffmedianhourlypercent_tag,
COUNT(1) AS number_of_companies
FROM gender_pay_gap_21_22
GROUP BY 1
```

	diffmedianhourlypercent_tag	number_of_companies
	text	bigint
1	Has data	10174

--Check for diffmeanbonuspercent

```
SELECT CASE
    WHEN diffmeanbonuspercent IS NULL THEN 'Missing Data'
    ELSE 'Has data'
END AS diffmeanbonuspercent_tag,
COUNT(1) AS number_of_companies
FROM gender_pay_gap_21_22
GROUP BY 1
```

	diffmeanbonuspercent_tag	number_of_companies
	text	bigint
1	Has data	10174

--Check for diffmedianbonuspercent

```
SELECT CASE
    WHEN diffmedianbonuspercent IS NULL THEN 'Missing Data'
    ELSE 'Has data'
END AS diffmedianbonuspercent_tag,
COUNT(1) AS number_of_companies
FROM gender_pay_gap_21_22
GROUP BY 1
```

	diffmedianbonuspercent_tag	number_of_companies
	text	bigint
1	Has data	10174

--Check for malebonuspercent

```
SELECT CASE
    WHEN malebonuspercent IS NULL THEN 'Missing Data'
    ELSE 'Has data'
END AS malebonuspercent_tag,
COUNT(1) AS number_of_companies
FROM gender_pay_gap_21_22
GROUP BY 1
```

	malebonuspercent_tag	number_of_companies
	text	bigint
1	Has data	10174

--Check for femalebonuspercent

```
SELECT CASE
    WHEN femalebonuspercent IS NULL THEN 'Missing Data'
    ELSE 'Has data'
END AS femalebonuspercent_tag,
COUNT(1) AS number_of_companies
FROM gender_pay_gap_21_22
GROUP BY 1
```

	femalebonuspercent_tag text	number_of_companies bigint
1	Has data	10174

--Check for malelowerquartile

```
SELECT CASE
    WHEN malelowerquartile IS NULL THEN 'Missing Data'
    ELSE 'Has data'
END AS malelowerquartile_tag,
COUNT(1) AS number_of_companies
FROM gender_pay_gap_21_22
GROUP BY 1
```

	malelowerquartile_tag text	number_of_companies bigint
1	Has data	10174

--Check for femalelowerquartile

```
SELECT CASE
    WHEN femalelowerquartile IS NULL THEN 'Missing Data'
    ELSE 'Has data'
END AS femalelowerquartile_tag,
COUNT(1) AS number_of_companies
FROM gender_pay_gap_21_22
GROUP BY 1
```

	femalelowerquartile_tag text	number_of_companies bigint
1	Has data	10174

--Check for malelowermiddlequartile

```
SELECT CASE
    WHEN malelowermiddlequartile IS NULL THEN 'Missing Data'
    ELSE 'Has data'
END AS malelowermiddlequartile_tag,
COUNT(1) AS number_of_companies
FROM gender_pay_gap_21_22
GROUP BY 1
```

	malelowermiddlequartile_tag text	number_of_companies bigint
1	Has data	10174

--Check for femalelowermiddlequartile

```
SELECT CASE
    WHEN femalelowermiddlequartile IS NULL THEN 'Missing Data'
    ELSE 'Has data'
END AS femalelowermiddlequartile_tag,
COUNT(1) AS number_of_companies
FROM gender_pay_gap_21_22
GROUP BY 1
```

	femalelowermiddlequartile_tag 	number_of_companies 
	text	bigint
1	Has data	10174

--Check for maleuppermiddlequartile

```
SELECT CASE
    WHEN maleuppermiddlequartile IS NULL THEN 'Missing Data'
    ELSE 'Has data'
END AS maleuppermiddlequartile_tag,
COUNT(1) AS number_of_companies
FROM gender_pay_gap_21_22
GROUP BY 1
```

	maleuppermiddlequartile_tag 	number_of_companies 
	text	bigint
1	Has data	10174

--Check for femaleuppermiddlequartile

```
SELECT CASE
    WHEN femaleuppermiddlequartile IS NULL THEN 'Missing Data'
    ELSE 'Has data'
END AS femaleuppermiddlequartile_tag,
COUNT(1) AS number_of_companies
FROM gender_pay_gap_21_22
GROUP BY 1
```

	femaleuppermiddlequartile_tag 	number_of_companies 
	text	bigint
1	Has data	10174

--Check for maletopquartile

```
SELECT CASE
    WHEN maletopquartile IS NULL THEN 'Missing Data'
    ELSE 'Has data'
END AS maletopquartile_tag,
COUNT(1) AS number_of_companies
FROM gender_pay_gap_21_22
GROUP BY 1
```

	maletopquartile_tag 	number_of_companies 
	text	bigint
1	Has data	10174

--Check for femaletopquartile

```
SELECT CASE
    WHEN femaletopquartile IS NULL THEN 'Missing Data'
    ELSE 'Has data'
END AS femaletopquartile_tag,
COUNT(1) AS number_of_companies
FROM gender_pay_gap_21_22
GROUP BY 1
```

	femaletopquartile_tag text	number_of_companies bigint
1	Has data	10174

THE UK-WIDE PAY GAP

5. Choose which column you will use to calculate the pay gap. Will you use `DiffMeanHourlyPercent` or `DiffMedianHourlyPercent`? Can you justify your choice?

Answer: I would use the `DiffMedianHourlyPercent` as the primary metric to calculate the gender pay gap because it is less affected by outliers and extreme values and provides a better representation of what a typical employee earns.

In this dataset, the mean hourly pay gap reflects the average difference in pay and can be influenced by extreme values, such as highly paid executives. In contrast, the median represents the middle value of hourly pay when employees are arranged in order, making it less sensitive to outliers and more representative of the typical employee experience.

Therefore, the median answers the question: *“What does a typical woman earn compared to a typical man?”*

In addition to the `DiffMedianHourlyPercent`, I would also complement the analysis with mean hourly pay gap (`DiffMeanHourlyPercent`) to assess the impact of high earners, as well as quartile distribution analysis to understand how men and women are represented across different pay levels. This provides a more complete view of both the magnitude, and the underlying drivers of the gender pay gap.

This is particularly important in this dataset, where the presence of skewness and extreme values makes the median a more reliable measure of central tendency.

6. Use an appropriate metric to find the average gender pay gap across all the companies in the data set. Did you use the mean or the median as your averaging metric? Can you justify your choice?

Answer: I would use the `DiffMedianHourlyPercent` as the primary metric to calculate the gender pay gap because it is less affected by outliers and extreme values and provides a better representation of what a typical employee earns.

Between hourly pay and bonus pay, hourly pay is more stable than bonus pay as it is the basic and consistent form of paying employees.

To compute the average gender pay gap across companies, I used the **median of the median hourly pay gap**, `DiffMedianHourlyPercent` as the primary metric.

Based on the dataset, the **mean across companies is 12.31%**, while the **median is lower at 9.8%**, indicating that the distribution is skewed. This is further supported by the presence of extreme values, with a minimum of -275.9% and a maximum of 100%.

The percentile distribution also shows a wide spread, with the middle 50% of companies ranging from 1% to 21.7%, and the top 10% reaching as high as 33.4%. These results confirm that a small number of extreme values are influencing the mean.

Therefore, the **median is a more appropriate measure**, as it is less sensitive to outliers and better represents the typical company's gender pay gap.

--Compute the mean, median, min and max of `diffmedianhourlypercent`

```
SELECT
    AVG(diffmedianhourlypercent) AS mean_across_companies,
    PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY
diffmedianhourlypercent) AS median_across_companies,
    MIN(diffmedianhourlypercent) AS min_diffmedianhourlypercent,
    MAX(diffmedianhourlypercent) AS max_diffmedianhourlypercent
FROM gender_pay_gap_21_22;
```

	mean_across_companies numeric	median_across_companies double precision	min_diffmedianhourlypercent numeric	max_diffmedianhourlypercent numeric
1	12.3131708275997641	9.8	-275.9	100

--Compute the standard deviation (percentile distribution) of `diffmedianhourlypercent`

```
SELECT
    PERCENTILE_CONT(0.1) WITHIN GROUP (ORDER BY
diffmedianhourlypercent) AS p10,
    PERCENTILE_CONT(0.25) WITHIN GROUP (ORDER BY
diffmedianhourlypercent) AS p25,
    PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY
diffmedianhourlypercent) AS p50,
    PERCENTILE_CONT(0.75) WITHIN GROUP (ORDER BY
diffmedianhourlypercent) AS p75,
```

```
PERCENTILE_CONT(0.9) WITHIN GROUP (ORDER BY
diffmedianhourlypercent) AS p90
FROM gender_pay_gap_21_22
WHERE diffmedianhourlypercent IS NOT NULL;
```

	p10 double precision	p25 double precision	p50 double precision	p75 double precision	p90 double precision
1	-2	1	9.8	21.7	33.4

7. What are some caveats we need to be aware of when reporting the figure we've just calculated?

Answer: While the median of `DiffMedianHourlyPercent` provides a **reliable and useful indication** of the typical gender pay gap across companies, it has several limitations that should be considered.

The median of medians reflects the *“typical company”* rather than the true overall pay gap across all employees. It does not capture pay distribution within roles, differences by job level or leadership imbalances.

First, it reflects the **“typical company” rather than the true overall pay gap across all employees**. It summarises each company into a single value and therefore does not capture differences within organisations, such as variations by role, job level, or leadership representation.

Second, the metric is based only on **full-pay relevant employees**, excluding those on reduced pay due to leave (such as maternity, paternity, sick, sabbatical, or other forms of leave). As these groups are more likely to include women, this may understate the gender pay gap.

Third, it does not take into account **company size**, meaning smaller and larger employers are given equal weight. As a result, the figure may not fully reflect the experience of the workforce as a whole.

In addition, while the metric tells us **how large the gender pay gap is, which 9.8% in this dataset**, it **does not explain why the gap exists**. Understanding the underlying causes requires further analysis, such as examining how men and women are distributed across different pay levels.

Finally, the data is **self-reported by employers** and reflects a single point in time. While reporting follows standard guidelines, differences in interpretation and reporting practices may introduce some inconsistencies. An example is determining if a pay is an allowance or an expense, which may potentially affect the hourly pay rate of the company.

Therefore, this measure should be seen as a **reliable but partial view**, and it is important to consider it alongside other information to gain a fuller understanding of gender pay differences.

LARGEST PAY GAPS

8. What are the 10 companies with the largest pay gaps skewed towards men?

Answer: Top 10 companies with the largest pay gaps skewed towards men:

1. HPI UK HOLDING LTD.
2. ATFC LIMITED
3. M. ANDERSON CONSTRUCTION LIMITED
4. PSJ FABRICATIONS LTD
5. HULL COLLABORATIVE ACADEMY TRUST
6. SERVICE INNOVATION GROUP-UK LIMITED
7. BRAND ENERGY & INFRASTRUCTURE SERVICES UK, LTD.
8. ROBINSON WEBSTER (HOLDINGS) LIMITED
9. THE LEARNING FOR LIFE PARTNERSHIP
10. GREENBROOK HEALTHCARE (HOUNSLOW) LIMITED

```
SELECT employername,
       companynumber,
       employerid,
       diffmedianhourlypercent,
       employersize,
       maleuppermiddlequartile,
       femaleuppermiddlequartile,
       maletopquartile,
       femaletopquartile
FROM gender_pay_gap_21_22
WHERE diffmedianhourlypercent IS NOT NULL
      AND diffmedianhourlypercent > 0
ORDER BY diffmedianhourlypercent DESC
LIMIT 10;
```

	employername character varying	companynumber character varying	employerid character varying	diffmedianhourlypercent numeric	employersize character varying	maleuppermiddlequartile numeric	femaleuppermiddlequartile numeric	maletopquartile numeric	femaletopquartile numeric
1	HPI UK HOLDING LTD.	05598207	6357	100	250 to 499	100	0	100	0
2	ATFC LIMITED	04712181	1557	100	250 to 499	0	0	0	0
3	M. ANDERSON CONSTRUCTION LIMITED	02207082	20315	100.0	250 to 499	100.0	0.0	100.0	0.0
4	PSJ FABRICATIONS LTD	01205595	19413	100	Less than 250	100	0	100	0
5	HULL COLLABORATIVE ACADEMY TRUST	08542806	6389	93	Not Provided	17	83	17	83
6	SERVICE INNOVATION GROUP-UK LIMITED	01443016	11051	90.4	250 to 499	0	0	0	0
7	BRAND ENERGY & INFRASTRUCTURE SERVICES UK, LTD.	08729615	2321	89.0	1000 to 4999	95.0	5.0	96.0	4.0
8	ROBINSON WEBSTER (HOLDINGS) LIMITED	01069599	10563	85.6	250 to 499	17.9	82.1	25.5	74.5
9	THE LEARNING FOR LIFE PARTNERSHIP	09675372	19375	82.6	Not Provided	4.7	95.3	18.8	81.2
10	GREENBROOK HEALTHCARE (HOUNSLOW) LIMITED	06025335	15080	77.1	500 to 999	18.3	81.7	40.9	59.1

Total rows: 10 Query complete 00:00:00.656

LF Ln 321, Col 1

9. What do you notice about the results? Are these well-known companies?

Answer: The top 10 companies with the largest gender pay gaps show **extreme values ranging from 77% to 100%**, and these gaps are closely linked to **male dominance in both the upper-middle and top pay quartiles**.

None of these companies are widely recognised household names in the UK or globally. Most appear to be smaller or specialised organisations, with many falling within the **250 to 499 employee size range**. A few are more recognisable within their sectors, such as *Brand Energy*

& Infrastructure Services UK Ltd and Greenbrook Healthcare (Hounslow) Limited, but they are not widely known public brands.

Several clear patterns emerge from the data:

First, complete male dominance at higher pay levels.

In companies such as *HPI UK Holding Ltd*, *M. Anderson Construction Limited*, and *PSJ Fabrications Ltd*, both the upper-middle and top pay quartiles are 100% male and 0% female. This indicates that **women have no representation from higher-paying roles, with no visible progression pipeline into senior positions.**

Second, strong male dominance with limited female representation.

For example, *Brand Energy & Infrastructure Services UK Ltd* shows approximately 95% male representation in both upper-middle and top quartiles, with only **minimal female presence.** This suggests that **gender imbalance begins earlier in the career pipeline and persists into senior roles.**

Third, mixed or female-leaning representation with still high pay gaps.

Companies such as *Robinson Webster (Holdings) Limited*, *The Learning for Life Partnership*, and *Greenbrook Healthcare (Hounslow) Limited* show relatively **higher female representation in upper-middle and top quartiles.** However, they still **exhibit large pay gaps** (approximately 77% to 86%), suggesting that **differences in pay structure** — such as variations in senior roles, pay levels, or bonus structures — may contribute to the gap.

Finally, potential data limitations or anomalies.

In companies such as *ATFC Limited* and *Service Innovation Group-UK Limited*, there appears to be no representation in the upper-middle and top quartiles. Given their relatively smaller workforce size, this may reflect data limitations, reporting issues, or highly unusual workforce structures, and should be interpreted with caution.

Overall, these findings suggest that the most extreme gender pay gaps are largely driven by **how men and women are distributed across different roles and pay levels, particularly in higher-paying positions, particularly the absence or underrepresentation of women in higher-paying roles.** However, these companies are likely to be **outliers** and may not be representative of the broader economy.

10. Apply some additional filtering to pick out the most significant companies with large pay gaps.

Answer: First, to identify the most significant cases, I focused on **larger organisations** and selected those with a median hourly pay gap above the overall median, while excluding very extreme values. This helps ensure the results **reflect realistic workforce structures** rather than outliers.

Next, I ensured that **both men and women are represented in the top pay quartile**, so that the results reflect genuine differences in higher pay levels rather than situations where one gender is absent. This allows us to focus on companies where the pay gap is **both meaningful and impactful.**

Finally, I included a condition on the **upper-middle pay quartile** to assess whether the imbalance begins earlier in the pay structure. This helps identify a **potential career pipeline issue**, where differences in representation start before the most senior roles.

```
SELECT
    employername,
    companynumber,
    employerid,
    employersize,
    diffmedianhourlypercent,
    maletopquartile,
    femaletopquartile,
    maleuppermiddlequartile,
    femaleuppermiddlequartile
FROM gender_pay_gap_21_22
WHERE diffmedianhourlypercent BETWEEN 9.8 AND 80 --more than the
median and excluded extreme or anomalous values
    AND employersize IN ('5,000 to 19,999', '20,000 or more') --
larger organisations, more impact
    AND maletopquartile >= 60 --Focus on structural imbalance where
men dominate higher-paying roles
    AND femaletopquartile > 0 --Ensure both genders are present
    AND maleuppermiddlequartile >= 60 --Progression pipeline to
senior level
ORDER BY diffmedianhourlypercent DESC
LIMIT 10;
```

Showing rows: 1 to 8 Page No: 1 of 1									
	employername character varying	companynumber character varying	employerid character varying	employersize character varying	diffmedianhourlypercent numeric	maletopquartile numeric	femaletopquartile numeric	maleuppermiddlequartile numeric	femaleuppermiddlequartile numeric
1	J D WETHERSPOON PLC	01709784	13691	20,000 or more	29.5	71	29	67	33
2	BARCLAYS EXECUTION SERVICES LIMITED	01767980	1808	20,000 or more	27.1	69.0	31.0	64.0	36.0
3	WPP 2005 LIMITED	01003653	14266	20,000 or more	23.6	66	34	68	32
4	BRITISH AIRWAYS PLC	01777777	549	20,000 or more	22.2	81.5	18.5	68.6	31.4
5	BRITISH TELECOMMUNICATIONS PUBLIC LIMITED COMPANY	01800000	2460	20,000 or more	12.9	74.4	25.6	74.7	25.3
6	NETWORK RAIL INFRASTRUCTURE LIMITED	02904587	8877	20,000 or more	12.9	86.1	13.9	85.6	14.4
7	GXO LOGISTICS UK LIMITED	SC037270	13994	20,000 or more	12.4	86.6	13.4	80.5	19.5
8	DHL SERVICES LIMITED	03324664	4099	20,000 or more	10	82	18	85	15

Total rows: 8 Query complete 00:00:00.719 LF Ln 359, Col 1

11. How would you report on the results? Can we say that these companies are engaging in unlawful pay discrimination?

Answer: The results show that even among well-known and large organisations with over 20,000 employees, **gender pay gaps remain significant**, ranging from approximately 10% to nearly 30%. These gaps are consistently associated with the **underrepresentation of women in higher-paying roles**, as shown by the high proportion of men in both the upper-middle and top pay quartiles.

This suggests that the gender pay gap is driven not only by differences at the most senior level, but also by a **broader career pipeline issue**, where imbalances begin earlier in progression into higher-paying roles. Given the size of these organisations, the impact of these structural imbalances is likely to be substantial.

However, it is important to note that a gender pay gap does not necessarily indicate **unlawful pay discrimination**. The data reflects differences in overall pay between men and women,

which are often influenced by how employees are distributed across roles and levels within an organisation. Based on the available data, the gaps appear to be linked to differences in representation rather than unequal pay for the same work.

Addressing the gap therefore requires **improving representation not only at the most senior levels, but across the progression pipeline into higher-paying roles.**

GEOGRAPHIC DIFFERENCES

12. What's the average pay gap in London versus outside London?

Answer: Companies were classified as London or outside London based on their postcode. Employers were identified as London-based if their postcode began with **standard UK London area prefixes (E, EC, N, NW, SE, SW, W, WC)**, which broadly represent East, West, North, South, and Central London. While this approach provides a reasonable proxy for location, it is not fully precise and reflects the **employer's registered office rather than the actual distribution of employees.**

The analysis shows that the **median gender pay gap is higher in London (11%) than outside London (9.3%)**, indicating that pay differences are slightly more pronounced among London-based organisations. However, the difference is relatively modest, suggesting that the **gender pay gap is present across all regions.**

The higher gap in London may reflect the concentration of high-paying industries and roles, where men are more likely to be represented in senior positions.

Overall, this indicates that regional location alone does not fully explain the gender pay gap. Instead, **the gap is more strongly influenced by workforce composition**, particularly the distribution of men and women across different pay levels.

```
WITH location_indicator AS (  
  SELECT  
    employername,  
    postcode,  
    diffmedianhourlypercent,  
    maletopquartile,  
    femaletopquartile,  
    maleuppermiddlequartile,  
    femaleuppermiddlequartile,  
    CASE  
      WHEN postcode ~* '^(E|EC|N|NW|SE|SW|W|WC)' THEN 'London'  
      --matches regular expression, case-insensitive  
      ELSE 'Outside London'  
    END AS region  
  FROM gender_pay_gap_21_22  
  WHERE diffmedianhourlypercent IS NOT NULL  
    AND maletopquartile IS NOT NULL  
    AND femaletopquartile IS NOT NULL  
    AND maleuppermiddlequartile IS NOT NULL  
    AND femaleuppermiddlequartile IS NOT NULL  
)
```

```

SELECT
    region,
    COUNT(*) AS number_of_companies,
    PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY
diffmedianhourlypercent) AS median_gap, --median per region
    PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY maletopquartile) AS
median_maletopquartile,
    PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY femaletopquartile)
AS median_femaletopquartile,
    PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY
maleuppermiddlequartile) AS median_maleuppermiddlequartile,
    PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY
femaleuppermiddlequartile) AS median_femaleuppermiddlequartile
FROM location_indicator
GROUP BY region
ORDER BY 1;

```

	region text	number_of_companies bigint	median_gap double precision	median_maletopquartile double precision	median_femaletopquartile double precision	median_maleuppermiddlequartile double precision	median_femaleuppermiddlequartile double precision
1	London	3457	11	61.7	37	53.4	45.3
2	Outside London	6717	9.3	60	38.9	51.9	46.5

13. What's the average pay gap in London versus Birmingham?

Answer: The analysis shows that the **median gender pay gap is higher in London (11%) compared to Birmingham (9.1%)**, indicating that gender pay differences are slightly more pronounced among London-based organisations.

However, the difference is relatively modest, suggesting that the **gender pay gap is present in both locations**. The higher gap in London may be influenced by the concentration of high-paying industries such as finance, technology and professional services and high-paying roles, where men are more likely to be represented in senior positions.

In contrast, Birmingham has a more balanced regional economy with fewer very high-paying roles and a narrower pay distribution. This can result in a slightly lower gender pay gap, although the gap remains present.

Overall, this suggests that location alone does not fully explain the gender pay gap, which is **more strongly influenced by workforce composition** and representation across different pay levels.

```

WITH location_indicator AS (
    SELECT
        employername,
        postcode,
        diffmedianhourlypercent,
        maletopquartile,
        femaletopquartile,
        maleuppermiddlequartile,
        femaleuppermiddlequartile,
        CASE
            WHEN postcode ~* '^(E|EC|N|NW|SE|SW|W|WC)' THEN 'London'
            WHEN postcode ~* '^B' THEN 'Birmingham'
            ELSE 'Other'

```

```

        END AS region
    FROM gender_pay_gap_21_22
    WHERE diffmedianhourlypercent IS NOT NULL
        AND maletopquartile IS NOT NULL
        AND femaletopquartile IS NOT NULL
        AND maleuppermiddlequartile IS NOT NULL
        AND femaleuppermiddlequartile IS NOT NULL
)
SELECT
    region,
    COUNT(*) AS number_of_companies,
    PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY
diffmedianhourlypercent) AS median_gap, --median per region
    PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY maletopquartile) AS
median_maletopquartile,
    PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY femaletopquartile)
AS median_femaletopquartile,
    PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY
maleuppermiddlequartile) AS median_maleuppermiddlequartile,
    PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY
femaleuppermiddlequartile) AS median_femaleuppermiddlequartile
FROM location_indicator
WHERE region IN ('London', 'Birmingham')
GROUP BY region
ORDER BY region DESC;

```

	region text	number_of_companies bigint	median_gap double precision	median_maletopquartile double precision	median_femaletopquartile double precision	median_maleuppermiddlequartile double precision	median_femaleuppermiddlequartile double precision
1	London	3457	11	61.7	37	53.4	45.3
2	Birmingham	949	9.1	56	42	47.8	50

INDUSTRY DIFFERENCES

Reference:

<https://www.ons.gov.uk/businessindustryandtrade/business/activitysizeandlocation/adhocs/11697enterprisesbyuksic2007class>

14. What is the average pay gap within schools?

Answer: The analysis shows that the **median gender pay gap in the Education sector is approximately 26.6%**, which is significantly higher than the overall average of 9.8%.

This indicates that **a higher proportion of women in the workforce does not necessarily translate into a smaller pay gap**. Instead, the gap is likely driven by **differences in representation across roles and pay structures**, with women more concentrated in lower- and mid-level positions and men's pay are substantially higher than women in higher-paying leadership roles despite women having more representation,

This highlights that the gender pay gap is primarily influenced by **workforce structure** rather than overall gender representation.

```
SELECT
    COUNT(*) AS number_of_companies,
    PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY
diffmedianhourlypercent) AS median_hourlypay,
    PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY maletopquartile)
AS median_maletopquartile,
    PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY femaletopquartile)
AS median_femaletopquartile,
    PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY
maleuppermiddlequartile) AS median_maleuppermiddlequartile,
    PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY
femaleuppermiddlequartile) AS median_femaleuppermiddlequartile,
    PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY
malelowermiddlequartile) AS median_malelowermiddlequartile,
    PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY
femalelowermiddlequartile) AS median_femalelowermiddlequartile,
    PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY malelowerquartile)
AS median_malelowerquartile,
    PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY
femalelowerquartile) AS median_femalelowerquartile

FROM gender_pay_gap_21_22
WHERE LEFT(siccodes, 4) ILIKE '85%'
AND diffmedianhourlypercent IS NOT NULL
AND maletopquartile IS NOT NULL
AND femaletopquartile IS NOT NULL
AND maleuppermiddlequartile IS NOT NULL
AND femaleuppermiddlequartile IS NOT NULL

AND malelowermiddlequartile IS NOT NULL
AND femalelowermiddlequartile IS NOT NULL
AND malelowerquartile IS NOT NULL
```

AND femalelowerquartile IS NOT NULL;

	number_of_companies bigint	median_hourlypay double precision	median_maletopquartile double precision	median_femaletopquartile double precision	median_maleuppermiddlequartile double precision	median_femaleuppermiddlequartile double precision
1	942	26.55	33.7	65.95	26.4	73

maleuppermiddlequartile recision	median_femaleuppermiddlequartile double precision	median_malelowermiddlequartile double precision	median_femalelowermiddlequartile double precision	median_malelowerquartile double precision	median_femalelowerquartile double precision
26.4	73	20.6	79	15	84.5

Note: I filtered companies in the education sector using SIC codes starting with 85 and calculated the median gender pay gap across these organisations.

The caveat using this approach is that it only checks the first code and excludes companies where Education sector is not the first listed SIC. Upon checking the dataset further, all companies under the Education sector has the SIC Code related to Education listed as the first code. Therefore, the query was able to extract all companies under the Education sector.

The query below resulted to having 4 additional records having '85' in the siccode in the middle which further confirms that the above query suffices to generate the complete set of companies in the Education sector.

```
SELECT siccodes, COUNT(1)
FROM gender_pay_gap_21_22
WHERE LEFT(siccodes, 4) ILIKE '%85%'
GROUP BY 1
ORDER BY 1
```

15. What is the average pay gap within banks?

Answer: The analysis shows that the **median gender pay gap in the Finance and Insurance sector is approximately 23.5%**, which is significantly higher than the overall median. This suggests that the **gender pay gap is more pronounced in high-paying sectors**. The gap is likely driven by the concentration of men in higher-paying roles and the presence of bonus-driven compensation structures, which can widen overall pay differences.

Similar to the Education sector, this highlights that the gender pay gap is primarily influenced by **workforce composition and progression into senior roles**, rather than overall gender representation.

```
SELECT
    COUNT(*) AS number_of_companies,
    PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY
diffmedianhourlypercent) AS median_hourlypay,
    PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY maletopquartile)
AS median_maletopquartile,
    PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY femaletopquartile)
AS median_femaletopquartile,
    PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY
maleuppermiddlequartile) AS median_maleuppermiddlequartile,
    PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY
femaleuppermiddlequartile) AS median_femaleuppermiddlequartile
FROM gender_pay_gap_21_22
WHERE LEFT(siccodes, 4) ILIKE '64%' OR LEFT(siccodes, 4) ILIKE '66%'
```

```

AND diffmedianhourlypercent IS NOT NULL
AND maletopquartile IS NOT NULL
AND femaletopquartile IS NOT NULL
AND maleuppermiddlequartile IS NOT NULL
AND femaleuppermiddlequartile IS NOT NULL;

```

	number_of_companies bigint	median_hourlypay double precision	median_maletopquartile double precision	median_femaletopquartile double precision	median_maleuppermiddlequartile double precision	median_femaleuppermiddlequartile double precision
1	362	23.450000000000003	71.300000000000001	28.700000000000003	58.95	41.05

Note: I filtered companies in the education sector using SIC codes starting with 64 and 66 and calculated the median gender pay gap across these organisations. The caveat using this approach is that it only checks the first code and excludes companies where Finance & insurance sector is not the first listed SIC.

The query below resulted to having 59 additional records having '64' or '66' in the siccode in the middle which further confirms that the above query suffices to generate the complete set of companies in the Education sector.

```

SELECT siccodes, COUNT(1)
FROM gender_pay_gap_21_22
WHERE LEFT(siccodes, 4) ILIKE '%64%' OR LEFT(siccodes, 4) ILIKE
'%66%'
AND diffmedianhourlypercent IS NOT NULL
GROUP BY 1
ORDER BY 1;

```

16. Is there a relationship between the number of employees at a company and the average pay gap?

Answer: There is no clear or consistent relationship between company size and gender pay gap. **While smaller companies may show more extreme values due to workforce size** (based on the results in Question 8 and 9), **large organisations can also exhibit significant pay gaps** (based on the results in Question 10 and 11).

Looking at the dataset as a whole, the **differences across company sizes are relatively small**. Median pay gaps range from approximately 7% to 11%, with the largest organisations (20,000+ employees) showing slightly lower gaps on average.

This suggests that company size alone is not a key driver of the gender pay gap. Instead, the gap is more likely influenced by **workforce composition, particularly the representation of men and women in higher-paying roles**.

```

SELECT
    employersize,
    COUNT(*) AS number_of_companies,
    PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY
diffmedianhourlypercent) AS median_gap
FROM gender_pay_gap_21_22
WHERE diffmedianhourlypercent IS NOT NULL
GROUP BY employersize
ORDER BY employersize;

```

	employersize character varying 🔒	number_of_companies bigint 🔒	median_gap double precision 🔒
1	1000 to 4999	2131	9
2	20,000 or more	62	7.15
3	250 to 499	4274	10.8
4	5000 to 19,999	464	9.2
5	500 to 999	2501	9.5
6	Less than 250	532	9.9
7	Not Provided	210	9.350000000000001
Total rows: 7 Query complete 00:00:00.655			

BONUS (Optional):

1. Can you find out what the 'SicCodes' column corresponds to? Is there a way we can understand what each SIC code represents? Search online for extra information.

Answer: The SicCodes refer to the Standard Industrial Classification. For this analysis, I used the UK SIC 2007 listing.

CLASSIFICATIONS

This dataset uses the 2007 revision to the Standard Industrial Classification (UK SIC 2007) in place of the 2003 revision Standard Industrial Classification (UK SIC 2003). The UK SIC 2007 is a major revision of UK SIC 2003 with changes at all levels of the SIC. Further details on Standard Industrial Classification can be found on the ONS

<http://www.ons.gov.uk/ons/guide-method/classifications/current-standard-classifications/index.html>

The broad industry group structure has been defined under UK SIC 2007 and is listed below:

Description	UK SIC Section	2007 Division
Agriculture, forestry & fishing	A	01/03
Production	B, C, D and E	05/39
Mining, quarrying & utilities	B, D and E	05/09, 35/39
Manufacturing	C	10/33
Construction	F	41/43
Wholesale and retail; repair of motor vehicles	G	45/47
Motor trades	G	45
Wholesale	G	46
Retail	G	47
Transport & storage (inc postal)	H	49/53
Accommodation & food services	I	55/56
Information & communication	J	58/63
Finance & insurance	K	64/66
Property	L	68
Professional, scientific & technical	M	69/75
Business administration and support services	N	77/82
Public administration & defence	O	84
Education	P	85
Health	Q	86/88
Arts, entertainment, recreation and other services	R, S, T and U	90/99

Reference:

<https://www.ons.gov.uk/businessindustryandtrade/business/activitysizeandlocation/adhocs/1697enterprisesbyuksic2007class>

Filename: ah634.xls

REFERENCES:

1. Information on how the data is collected
https://www.gov.uk/government/publications/gender-pay-gap-reporting-guidance-for-employers?_ga=2.132484942.150848908.1643707737-1164715231.1643022613
2. Data Dictionary of the gender_pay_gap_21_22 dataset
<https://gender-pay-gap.service.gov.uk/viewing/download>
3. Standard Industrial Classification UK SIC 2007 listing
<https://www.ons.gov.uk/businessindustryandtrade/business/activitysizeandlocation/adhocs/1697enterprisesbyuksic2007class>

Filename: ah634.xls